



Society of Petroleum Engineers

SPE-226626-MS

Experimental Study on the Impact of Multiple Sets of Natural Fractures on Hydraulic Fractures

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226626-MS](https://doi.org/10.2118/226626-MS)

This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

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Abstract

Shale reservoirs commonly exhibit well-developed bedding planes and multiple sets of natural fractures, resulting in complex and variable hydraulic fracture propagation paths. Previous research unveiled the interactions when the hydraulic fracture encounters a single natural fracture, however, in cases of multiple sets of natural fractures the interactions remain unclear. This study investigates the fracture propagation behaviors and fracture mechanisms of hydraulic fractures in shale with multiple natural fractures under different treatment conditions, explores the influence of multiple natural fractures on hydraulic fractures, and provides theoretical guidance for field hydraulic fracturing.

Using full-diameter shale cores with multiple natural fractures, triaxial experiments of varying bedding plane angles and hydraulic fracturing were conducted to analyze fracture propagation for diverse geological features and treatment parameters, clarify the impact of multiple natural fractures on hydraulic fractures, and identify fracturing mechanisms. Additionally, a 3D scanner was employed to extract fracture surface data, characterizing fracture morphology and surface roughness, and further analyzing post-fracturing fracture topography.

Study indicates that the compressive strength of shale with multiple sets of natural fractures decreases and then increases with bedding plane angles, with three stages ($0-30^\circ$, $30-60^\circ$, and $60-90^\circ$), and the strength varies significantly within the $0-30^\circ$ and $60-90^\circ$, which is different from shales of a single natural fracture. At 0° bedding angle, hydraulic fractures penetrate bedding planes to form straight fractures. As the bedding angle increases, fractures propagate stepwise along bedding planes, leading to increased surface roughness and complexity. This behavior aligns with stress state variations near bedding interfaces during fracture propagation. High-viscosity fluids and injection rates elevate net pressure within fractures, altering local stress states and creating complex fracture networks with stepwise bedding-aligned propagation, accompanied by shifts in failure mechanisms.

This study elucidates the mechanism behind strength variations in shale with multiple natural fractures as bedding angles change, revealing that localized stress state alteration near fractures is the primary

driver of propagation path deviations. Variations in geological conditions and treatment parameters modify stress states around fractures, leading to transitions of failure mechanisms. A novel fracture mechanism is proposed, where hydraulic fractures evolve from pure tensile failure to tensile-shear mixed failure.

Introduction

With the rapid development of the global economy, conventional oil and gas resources have become insufficient to meet developmental demands, making the exploitation of unconventional energy resources a focal point of international scholarly attention (Economides & Nolte, 1989; Wang et al., 2022; Li & Chang, 2021; Ma & Xie, 2018). As a critical supplement to conventional energy, shale oil and gas have garnered significant interest due to their abundant reserves and substantial development potential (Zeng et al., 2022; Zou et al., 2021; Zhao et al., 2019). However, shale reservoirs are characterized by low porosity and permeability, posing considerable challenges for hydrocarbon extraction (Liu et al., 2018; Bao et al., 2016; He et al., 2023). Hydraulic fracturing, as a vital stimulation technology for unconventional tight reservoirs (Zhang et al., 2019; Xu et al., 2024), can effectively enhance shale oil and gas recovery. Nevertheless, the widespread development of bedding planes and multiple sets of natural fractures in shale reservoirs, coupled with variations in bedding cementation strength, tensile strength, and internal friction angles, result in complex and variable hydraulic fractures propagation paths. Furthermore, field monitoring struggles to capture the dynamic propagation process of hydraulic fractures, making precise control of fracture morphology challenging. Therefore, elucidating the initiation and propagation mechanisms of hydraulic fractures in shale reservoirs is imperative.

Current research on hydraulic fractures propagation in shale primarily employs numerical simulation methods. The finite element method has been applied to investigate fracture propagation within (Luo et al., 2023) and between (Suo et al., 2022) bedding layers. (Zhai et al., 2020) investigated the influence of cohesive material properties such as bedding planes and random natural fractures on hydraulic fracturing parameters through the cohesive element method. (Zhou et al., 2020) analyzed the impact of bedding density, bedding strength, and fracturing engineering parameters on the propagation law of hydraulic fractures by using the discrete element method. Additional approaches include phase-field methods (Lu et al., 2023) and boundary element methods (Zhou et al., 2020). However, existing numerical simulations typically represent bedding planes by inserting cohesive elements into reservoir rock models, failing to accurately reflect the arbitrary propagation paths of hydraulic fractures in the presence of multiple natural fracture sets.

Physical experimentation remains the most direct method for observing fracturing phenomena, with true triaxial experiments capable of replicating subsurface rock stress conditions. Current research achievements in shale hydraulic fracturing experiments include: Li et al. (2019) identified that shale fracturing pressure relates to bedding dip angles; Fu et al. (2021) observed three interaction outcomes (penetration, arrest, and diversion) when hydraulic fractures encounter bedding planes, highlighting this as the primary mechanism for complex fracture network formation; Chang et al. (2020) and Zheng et al. (2022) identified two propagation modes post bedding plane interaction: direct penetration or bedding capture; Meng et al. (2021), Zhai et al. (2022), and Liu et al. (2023) demonstrated significant bedding influences on mechanical properties and failure modes. However, these scholars did not conduct in-depth analysis on how bedding and natural fractures affect hydraulic fractures.

Previous studies have investigated the interaction mechanisms between hydraulic fractures and natural fractures. Anderson (1981) identified the frictional coefficient of natural fracture surfaces and normal stress as primary parameters influencing whether hydraulic fractures can penetrate through natural fractures. Teufel and Clark (1984) discovered that the horizontal minimum principal stress also affects the penetration capability of hydraulic fractures through natural fractures. Renshaw and Pollard (1995) established critical conditions for orthogonal intersection between hydraulic fractures and natural fractures. However, in actual reservoirs containing multiple sets of natural fractures, hydraulic fractures propagation is simultaneously

influenced by operational parameters. Existing studies have not thoroughly elucidated the mechanisms by which multiple natural fracture groups affect hydraulic fractures development of varying operational conditions.

To investigate the influence of multiple groups of natural fractures on hydraulic fractures, this study employs full-diameter core samples from real reservoirs to conduct laboratory hydraulic fracturing experiments. Through 3D fracture surface reconstruction, we investigate fracture propagation behaviors under various geological conditions and treatment parameters, clarifying the influence of multiple natural fractures sets of hydraulic fractures development and failure mechanisms. Additionally, 3D fracture surface data extraction enables quantitative analysis of post-fracturing surface morphology based on height parameters. The findings aim to provide practical guidance for shale oil and gas hydraulic fracturing operations.

Experimental Methodology and Similarity Analysis

This study employs full-diameter reservoir cores to conduct laboratory-scale hydraulic fracturing experiments utilizing an independently developed true triaxial fracturing system. Through similarity analysis, the experimental results partially replicate the perturbation effects of multiple natural fractures sets on hydraulic fractures propagation at the field scale.

Main Experimental Equipment

The experimental study utilized a large-scale true triaxial hydraulic fracturing system designed and established by the Rock Mechanics Laboratory at the China University of Petroleum (Beijing) to simulate hydraulic fractures propagation under various conditions. The system is shown in Fig. 1(a). The MTS-816 mechanical testing system (Fig. 1b) was employed to conduct triaxial compression experiments under different confining pressures and bedding orientations, aiming to investigate the influence of bedding angles on the mechanical properties of shale containing multiple natural fractures of varying confining pressures. Additionally, a SIMSCAN portable 3D scanner (Fig. 1c) based on 3D laser scanning technology was used to scan the hydraulic fracture surfaces after fracturing experiments. The obtained 3D fracture surface data were processed to extract fracture height parameters, enabling quantitative characterization and comparison of fracture morphology under different experimental conditions.

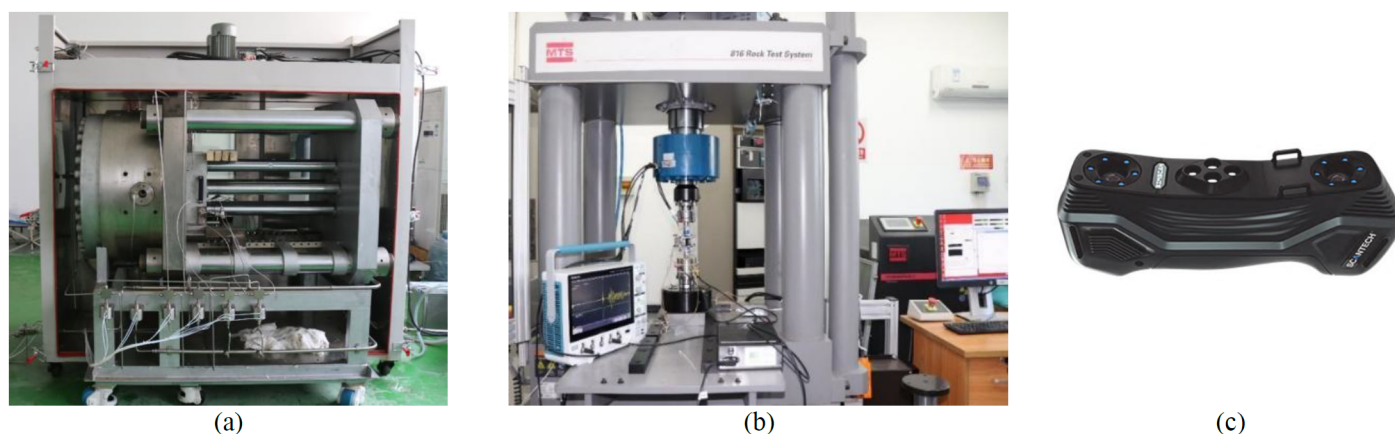


Figure 1—Experimental setups: (a) Large-scale true triaxial hydraulic fracturing system; (b) MTS-816 mechanical testing machine; (c) SIMSCAN portable 3D scanner.

Full-diameter shale cores extracted from a real reservoir were used for triaxial compression and hydraulic fracturing experiments. The full-diameter shale specimens were cylindrical with a diameter of 100 mm and a length of 200 mm. Smaller cores (25 mm in diameter and 50 mm in length) for triaxial compression tests

were drilled directly from the full-diameter cores. To meet the requirements of the hydraulic fracturing setup, the full-diameter cores were encapsulated with concrete into standard cubic specimens of 300 mm × 300 mm × 300 mm. During casting, the full-diameter core was positioned at the center of the specimen (Fig. 2a). Specimens with varying bedding angles were prepared by adjusting the inclination of the full-diameter core (Fig. 2b).

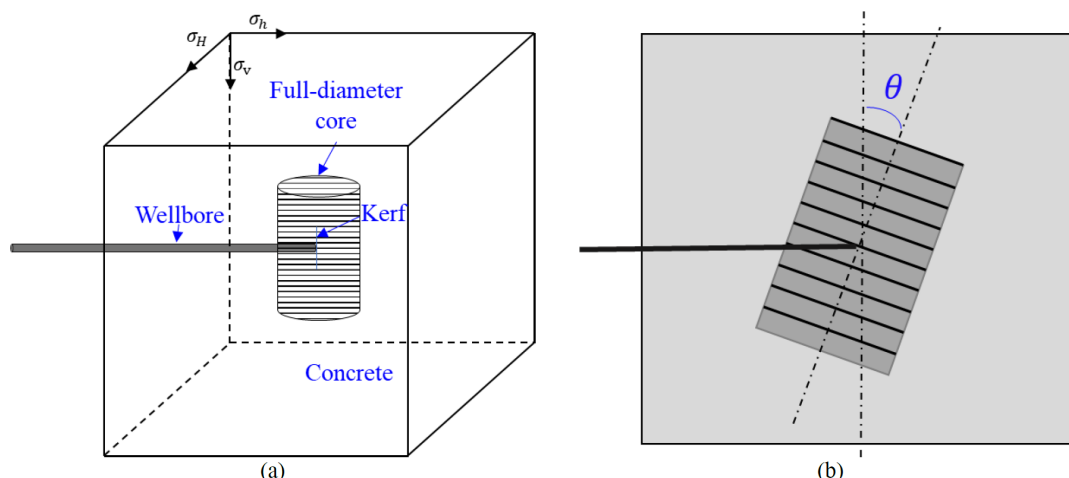


Figure 2—Schematic diagram of the specimen structure for the laboratory fracturing experiment(a) Full-diameter core location(b) Different bedding angles.

After curing for 15 days to ensure full consolidation, a vertical wellbore (150 mm in depth and 20 mm in diameter) was drilled at the center of each specimen to accommodate a simulated wellbore. The simulated wellbore, fabricated from metal tubing with an outer diameter of 3 mm and an inner diameter of 1.5 mm, had a total length of 250 mm. A 5-mm-deep circular notch was machined at the bottom of the wellbore, and a 15-mm open-hole section was reserved to facilitate fracture initiation. The annular space between the wellbore and the drilled hole was filled with epoxy resin adhesive. The specimens were allowed to cure at room temperature for 48 hours until the epoxy resin achieved full strength before conducting the fracturing experiments.

Experimental Scheme and Similarity Analysis

This study designed six groups of hydraulic fracturing comparative experiments to investigate the effects of bedding angle, fracturing fluid viscosity, and injection rate on shale hydraulic fractures propagation. A true triaxial loading system was used to simulate in-situ stress conditions, with triaxial stress parameters set according to actual shale reservoir stress states. The triaxial stresses applied during hydraulic fracturing experiments were maintained at 30 MPa, 25 MPa, and 20 MPa, the experimental parameters for each test group are detailed in Table 1. Guar gum solutions of varying concentrations served as fracturing fluids, with green fluorescent dye added to facilitate the observation of fracture morphology.

Table 1—Parameters of fracturing experiments

Specimen Number	Triaxial Stress σ_h - σ_H - σ_v /MPa	Viscosity /CP	Injection rate /(mL/min)	Bedding Angle /°
1	20-25-35	20	25	0
2	20-25-35	20	5	0
3	20-25-35	20	5	45
4	20-25-35	20	5	10
5	20-25-35	5	5	10

Specimen Number	Triaxial Stress σ_h - σ_H - σ_V /MPa	Viscosity /CP	Injection rate /(mL/min)	Bedding Angle /°
6	20-25-35	200	5	10

First, load the sample into the large-scale true triaxial physical simulation test system. Second, set the confining pressures and injection rates in three directions. After pre-compacting the sample, inject high-pressure fluid to induce fracturing. Finally, retrieve and section the sample. Based on the distribution of green fluorescent tracers, accurate and intuitive test results can be obtained. Combined with 3D scanning, the fracture morphology of the specimen can be precisely visualized, enabling subsequent analysis of fracture mechanisms and influencing factors of crack formation.

In the experimental design of this study, the applied triaxial stresses maintain the same hierarchical ordering as observed in field conditions: vertical stress (σ_V) > maximum horizontal principal stress (σ_H) > minimum horizontal principal stress (σ_h). Critically, the differential magnitudes among these triaxial stresses align proportionally with those of the target reservoir. Full-diameter reservoir cores containing multiple natural fracture sets were utilized, with each specimen centrally positioned within the loading assembly to mitigate boundary effects inherent to laboratory configurations. This approach effectively simulates field-scale hydraulic fractures propagation processes, thereby establishing a foundational basis for investigating the interference mechanisms between natural fracture networks and induced fractures.

Results and Analysis

Firstly, the mechanical properties of shale with multiple groups of natural fractures under different layering angles were analyzed. Secondly, by setting different experimental parameters, the hydraulic fracture propagation affected by natural fractures under different construction conditions was analyzed. The experimental parameters of each group and the corresponding fracture pressures are shown in Table 1. Specimens 1 to 2 change the liquid injection displacement to compare the influence of natural fractures on the fracture propagation law of shale under different displacement conditions; Specimens 2 to 4 change the bedding Angle to study the influence of natural fractures on the hydraulic fracture propagation behaviors under different stratum dip angles. Specimens 4 to 6 changed the viscosity of the fracturing fluid to study the influence of natural fractures under different viscosities on the hydraulic fracture propagation behaviors.

Mechanical Properties of Multi-Fractured Shale

This study first conducted triaxial compression experiments on shale core samples with different bedding angles (0°, 15°, 30°, 45°, 60°, 75°, 90°) of varying confining pressures (0, 10, 30, 50, 70 MPa) to analyze parameters such as elastic modulus, Poisson's ratio, compressive strength, and failure modes.

The experimental results for shale samples with multiple sets of natural fractures at different bedding angles are shown in Fig. 3. The compressive strength initially decreases and then increases with increasing bedding angle, exhibiting a three-stage distribution (0-30°, 30-60°, and 60-90°). At low bedding angles (0-30°), fractures are predominantly bedding plane fractures combined with inclined fractures, resulting in complex failure surfaces. At intermediate angles (45-60°), bedding plane failure is activated, while at high angles (75-90°), failure occurs as a single shear plane. Consequently, shale formations containing multiple natural fracture networks exhibit pronounced anisotropy, with bedding orientations governing their mechanical response.

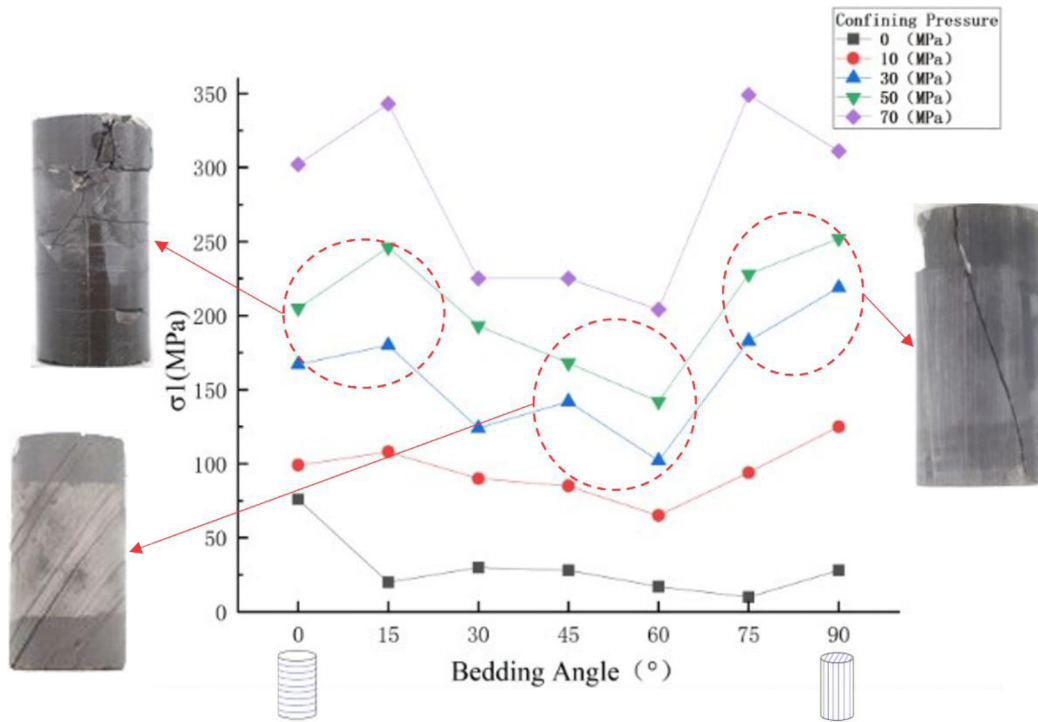


Figure 3—Relationship between σ_1 and bedding angle in shale.

Previously, scholars proposed the single structural plane theory to determine rock mass strength. As shown in Fig. 4a, when a single structural plane develops within the rock mass, assuming the angle between the maximum principal stress and the normal direction of the structural plane is β , Mohr's circle theory indicates that the normal stress and shear stress acting on the structural plane can be expressed as:

$$\begin{cases} \sigma = \frac{\sigma_1 + \sigma_3}{2} + \frac{\sigma_1 - \sigma_3}{2} \cos 2\beta \\ \tau = \frac{\sigma_1 - \sigma_3}{2} \sin 2\beta \end{cases} \quad (1)$$

Assuming the shear strength of the structural plane follows the Coulomb criterion

$$\tau = \sigma \tan \varphi + c \quad (2)$$

Where c is cohesion of the structural plane (MPa) and φ is the internal friction angle (°)

Substituting Equation (1) into Equation (2), the failure condition along the structural plane is derived as follows:

$$\sigma_1 - \sigma_3 = \frac{2(\sigma_3 \tan \varphi + c)}{(1 - \tan \varphi \cot \beta) \sin 2\beta} \quad (3)$$

The rock mass strength exhibits significant anisotropy controlled by β , as illustrated in Fig. 4b. When $\beta = 0$, the structural plane has no influence on rock mass strength, which approximates the intact rock strength. When $\beta = 45 + \varphi/2$, failure occurs along the structural plane, making the rock mass strength equal to the structural plane strength. When $\beta = 90$, the rock mass fails through lateral expansion along the weak plane, resulting in strength lower than that of intact rock.

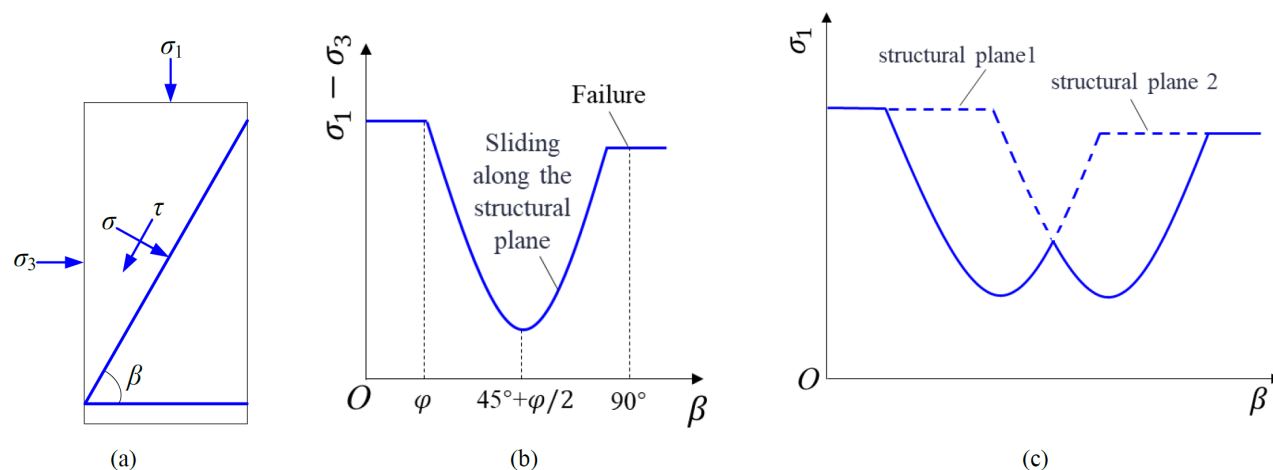


Figure 4—Strength of rock mass containing structural plane (a) Stress state (b) Strength curve of a single structural planes (c) Strength curve of two structural planes.

When two structural planes exist in the rock mass, the theory for a single structural plane and the superposition principle can be applied. The strength curves of each structural plane, when considered individually, are superimposed. The minimum envelope formed by these superimposed curves represents the strength curve of the rock mass containing two structural planes, as shown in Fig. 4c. For shale rock masses with multiple parallel structural planes of identical properties, the strength theory is similar to that for two structural planes. By superimposing the strength curves of each structural plane when considered individually, the strength curve of the rock mass with multiple structural planes is obtained. This allows for the determination of the strength of shale rock masses containing multiple sets of natural fractures of varying bedding angles, as illustrated in Fig. 5. The strength envelopes of three structural planes demonstrate strong agreement with the experimental data in Fig. 3, thereby explaining the variation pattern of strength with bedding angles for shale with multiple natural fractures.

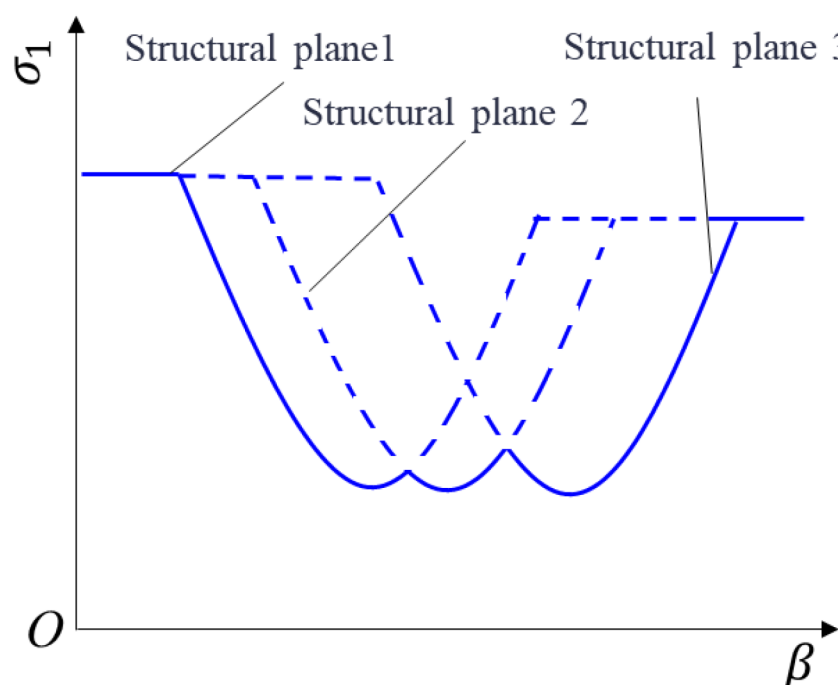


Figure 5—Strength curve of three structural planes.

Influence of Bedding on Fracture Propagation

Specimens 2-4 were subjected to shale fracturing experiments under identical displacement (5 mL/min) and viscosity (20 cp) conditions but varying bedding angles. The pressure curves under different bedding angles all exhibit distinct stages of fracture initiation and propagation extension, as shown in Fig. 6.

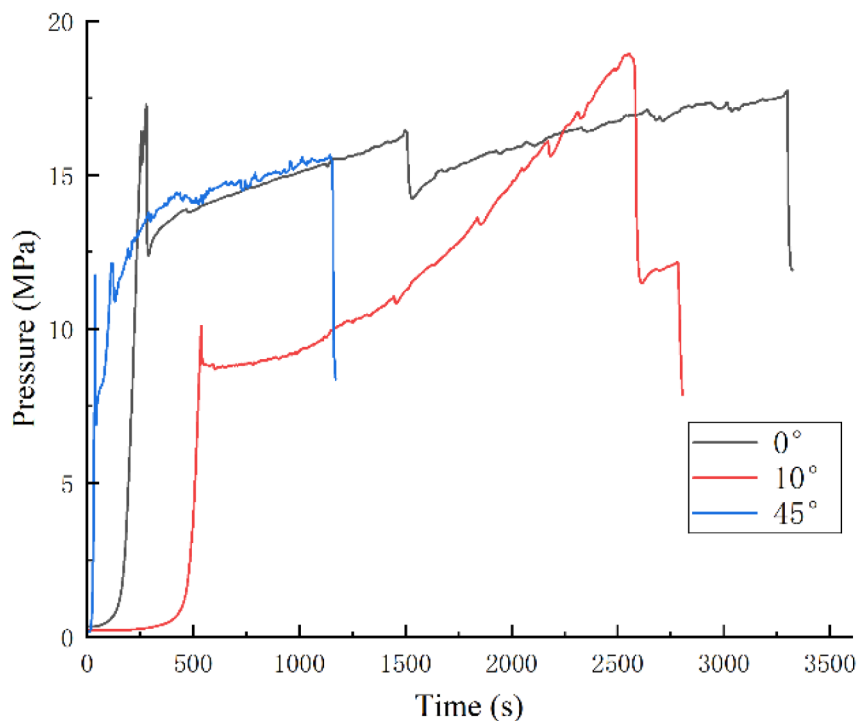


Figure 6—Injection pressure curves at different bedding angles.

After the experiment, Specimens 2-4 were sectioned to obtain fracture morphology diagrams. A SIMSCAN portable scanner was used to perform 3D scans of the fracture surfaces under different bedding angles. After processing, the morphology diagrams of hydraulic fractures (red) and induced bedding fractures (green) under different bedding angles were obtained, as shown in Fig. 7.

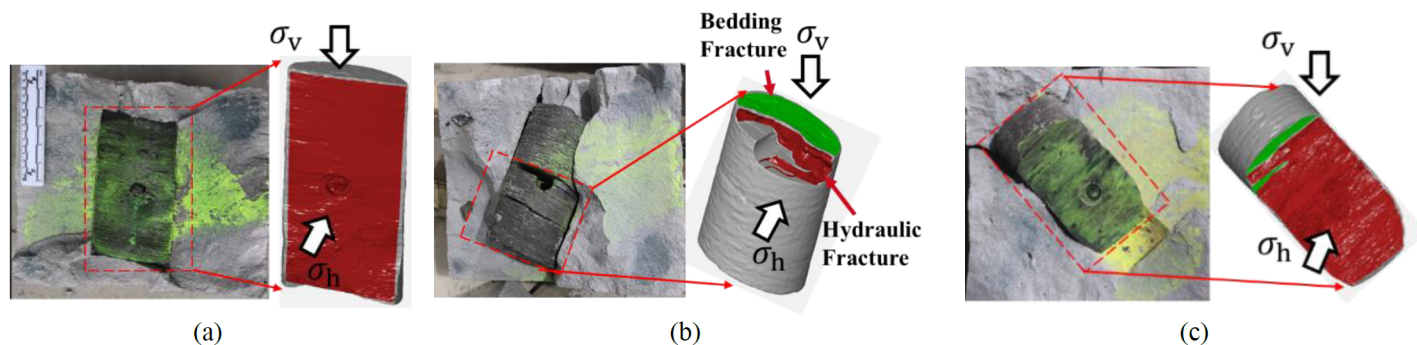


Figure 7—Fracture morphology under different bedding angles: (a) 0° bedding, (b) 10° bedding, (c) 45° bedding.

As shown in Fig. 7, at 0° bedding, the overall fracture surface is relatively flat, with fractures propagating primarily through trans-layer extension. At 10° bedding, the fracture surface becomes tortuous, and hydraulic fractures activate natural fractures to form bedding-parallel fractures, resulting in step-like propagation along the bedding planes. At 45° bedding, the fracture surface undergoes significant deflection, propagating in a step-like manner while further activating natural fractures to create a complex

fracture network. As evidenced by the compressive strength data for multi-fractured shale across bedding orientations in Fig. 3, specimens at 45° bedding exhibit significantly lower strength compared to those at 0°. This reduction directly contributes to altered fracture propagation paths, attributable to diminished mechanical competence at 45° orientations. When stratigraphic layers exhibit dip angles approaching 45°, natural fractures become structurally weak. Consequently, hydraulic fractures intersecting these natural fractures preferentially induce shear slippage rather than penetrating through, thereby altering propagation trajectories.

To quantitatively evaluate and compare fracture surface characteristics under different geological conditions, the three-dimensional height data of the fracture surfaces were extracted and statistically analyzed. Currently, many scholars categorize three-dimensional roughness characterization parameters into four types based on their geometric significance: dip angle type, area type, height type, and volumetric type. This study adopts the root mean square height (Z_{rms}), a volumetric-type parameter for characterizing the three-dimensional roughness of rock mass discontinuities, to describe the three-dimensional features of fracture surfaces. The root mean square height (Z_{rms}) is the standard deviation of height distribution, reflecting the dispersion of fracture surface height. It can be calculated as follows:

$$Z_{rms} = \left(\frac{1}{N} \iint z_{ij}^2 dx dy \right)^{1/2} \quad (4)$$

Where z_{ij} is the height (mm) of each data analysis point on the fracture surface; Z_{rms} is the root mean square height of the fracture surface (mm); N is the number of analysis points (dimensionless).

During the processing of 3D fracture surfaces, a coordinate plane parallel to the fracture surface is first established to facilitate direct extraction of height data. The geometric shapes and sizes of fracture surfaces vary across different hydraulic fracturing experimental results. To facilitate comparative analysis of height characteristics, this study adopts identical point spacing and analysis point counts for height data extraction. Characteristic diagrams and results of shale hydraulic fractures surfaces under different bedding angles, fracturing fluids, and viscosity conditions are shown in Fig. 8, while the Z_{rms} results are summarized in Table 2.

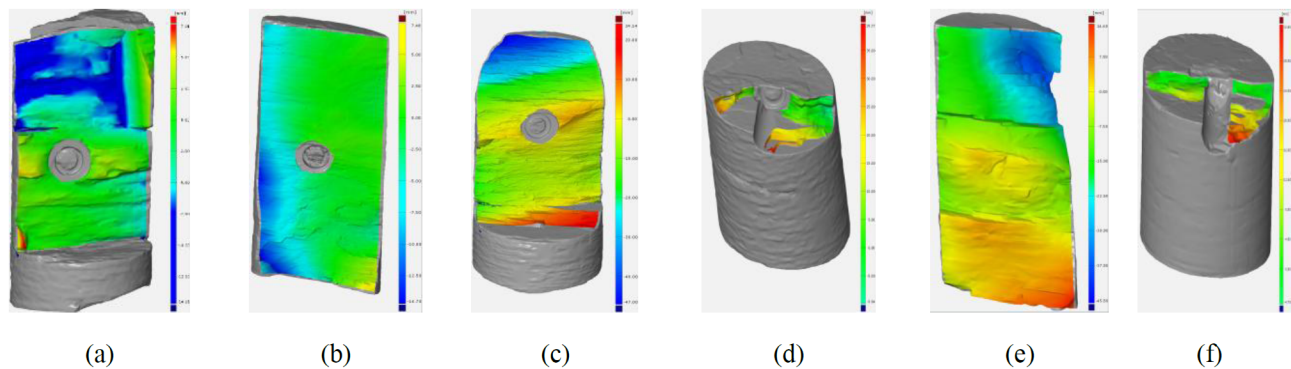


Figure 8—Crack surface height distribution of specimens 1# to 6#

Table 2—RMS height results of specimens 1# to 6#

Specimen Number	1	2	3	4	5	6
Z_{rms}/mm	5.1	3.8	12.3	11.9	11.6	9.5

Table 2 shows the height mean square error of fracture surfaces under different working conditions. From Fig. 7 and Table 2, it can be observed that under the same displacement (5 ml/min) and viscosity (20 cp), the fracture at 0° bedding angle exhibits bedding-transgressive propagation with lower surface roughness, resulting in a relatively smooth fracture surface. At 10° bedding angle, the fracture surface develops tortuous

paths and propagates stepwise along bedding planes, showing increased surface roughness. For 45° bedding angle, the fracture surface demonstrates greater deflection angles with stepwise propagation that terminates at bedding plane weaknesses.

The representative 3D height distribution maps of 0° and 45° bedding fracture surfaces were further processed to plot height variation curves along core length, as shown in Fig. 9b. Laser confocal scanning was performed on $1 \times 1 \text{ cm}^2$ areas of both 0° and 45° bedding hydraulic fracture surfaces, revealing micrometer-scale fracture morphology details in Fig. 9a and 9c respectively.

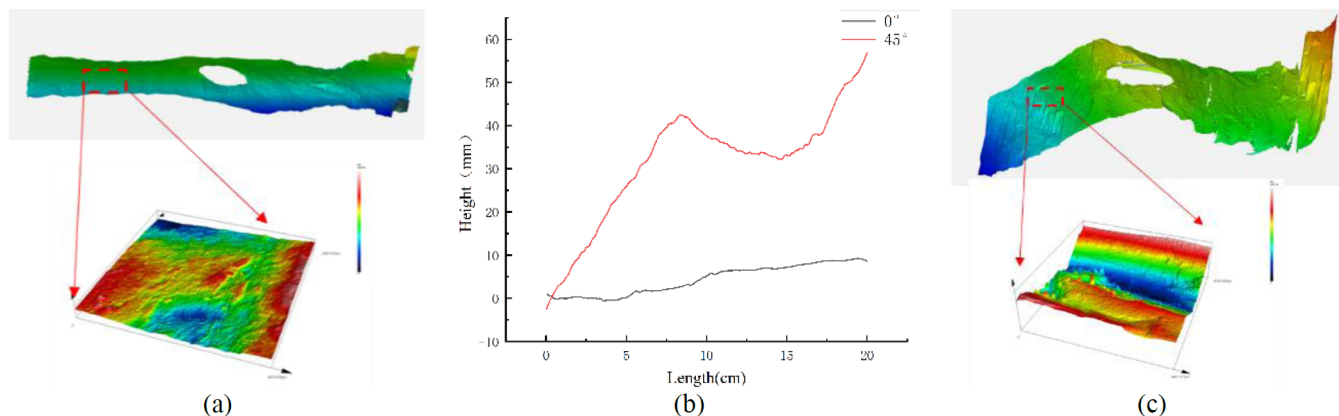


Figure 9—Variation in fracture surface elevation at different bedding angles

It is observed that the propagation paths of hydraulic fractures under 0° and 45° bedding angles undergo significant changes. Based on previous research, two fundamental criteria can be deduced: (1) Under idealized homogeneous conditions, fractures propagate along the direction of maximum tensile stress, and (2) the fracture propagation angle satisfies specific mechanical relationships. When layered shale exhibits a certain bedding angle, the propagation direction of hydraulic fractures along the bedding plane no longer remains perpendicular to the minimum principal stress but instead forms an angle θ , as shown in Fig. 10.

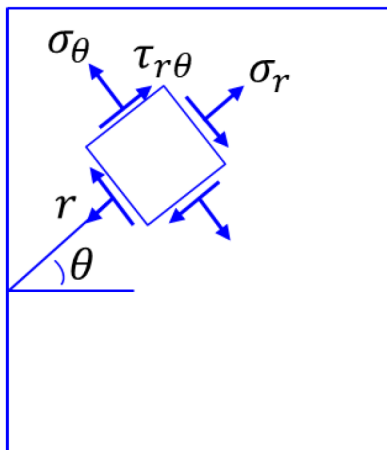


Figure 10—Crack tip stress components

In this scenario, the mechanical behavior of fracture propagation is governed by the combined action of normal and shear stresses, resulting in a mixed Mode I-II fracture. For plane problems, the stress components near the tip of a mixed-mode (I-II) fracture can be derived using the calculated stress fields at the tips of Mode I and Mode II cracks and the principle of superposition:

$$\sigma_r = \frac{1}{2\sqrt{2\pi r}} \left[K_I (3 - \cos\theta) \cos\frac{\theta}{2} + K_{II} (3\cos\theta - 1) \sin\frac{\theta}{2} \right] \quad (5)$$

$$\sigma_\theta = \frac{1}{2\sqrt{2\pi r}} \cos\frac{\theta}{2} [K_I (1 + \cos\theta) - 3K_{II} \sin\theta] \quad (6)$$

$$\tau_{r\theta} = \frac{1}{2\sqrt{2\pi r}} \cos\frac{\theta}{2} [K_I \sin\theta + K_{II} (3\cos\theta - 1)] \quad (7)$$

where, σ_r , σ_θ , and $\tau_{r\theta}$ is the three principal stresses in cylindrical coordinates (MPa); r is the distance from the crack tip (mm); K_I and K_{II} are the Mode I and Mode II fracture toughness values (MPa·m^{1/2}); θ is the angle between the crack propagation direction and deflection direction (°)

By differentiating Equation (4) with respect to θ and setting $\frac{\partial \sigma_\theta}{\partial \theta} = 0$, we derive the crack deflection angle equation. Notably, the solutions corresponding to $\cos\frac{\theta}{2} = 0$ ($\theta = \pm\pi$) are physically meaningless in this context, and are therefore disregarded. The resulting governing equation for crack deflection angle is expressed as:

$$K_I \sin\theta + K_{II} (3\cos\theta - 1) = 0 \quad (8)$$

Through Equations 4 to 7, it can be deduced that in complex fractures, the variation in local stress states at different fracture tips influences the direction of fracture propagation.

As clearly shown in Fig. 9, hydraulic fractures at 0° bedding angle exhibit through-bedding propagation with smooth fracture surfaces, indicating that the fracture mechanism is predominantly Mode I (tensile) failure. At 45° bedding angle, the fracture plane deflects and alters its propagation path. Combined with the compressive strength data of shale containing multiple structural planes at different bedding angles in Fig. 3, it is observed that the strength of shale with 45° bedding is lower than that of 0° bedding. This suggests that the primary reason for the altered propagation path lies in the inferior mechanical properties of 45° bedding shale. When fractures propagate to the bedding plane, changes in the surrounding stress state induce fracture path deviation, transforming hydraulic fractures from Mode I (tensile) failure to mixed tensile-shear (Mode I-II) composite failure. It can be concluded that there are mainly two types of hydraulic fracture propagation modes in shale reservoirs: the reservoir has no inclination Angle, the fracture initiation direction is perpendicular to the bedding, and the hydraulic fracture propagation is predominantly characterized by bedding-transgressive behavior. When the reservoir has a certain inclination Angle, hydraulic fractures deflect and expand in a stepped manner, as shown in Figure 11.

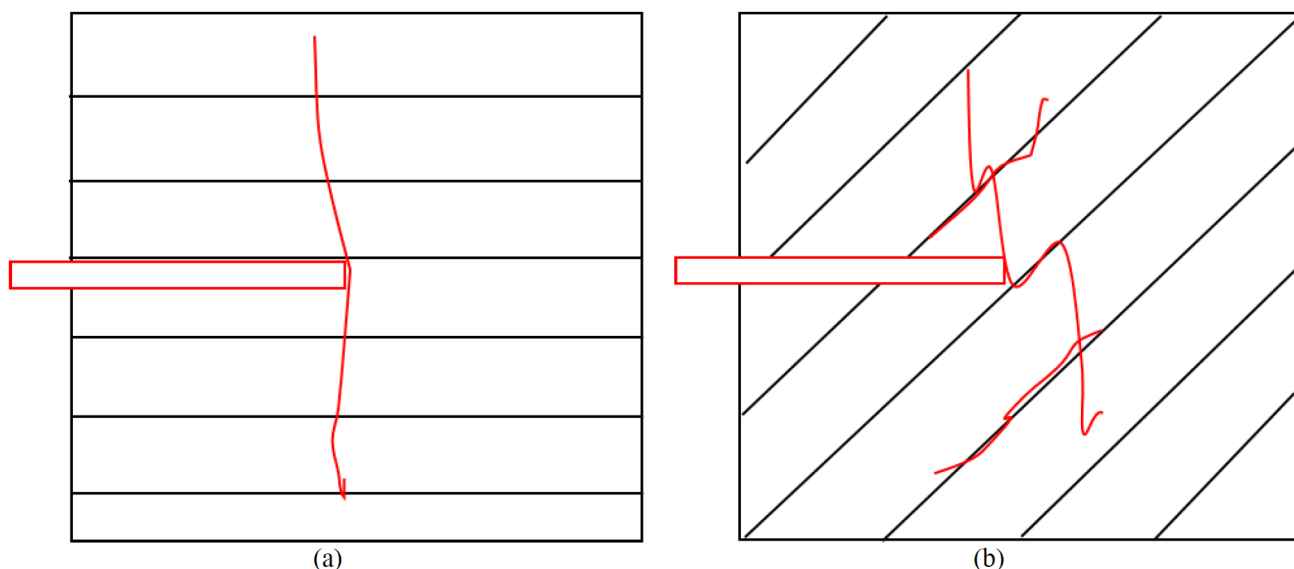


Figure 11—Crack propagation modes: (a) bedding-transgressive propagation; (b) deflection.

Influence of Injection rate and Viscosity on Fracture Propagation

Specimens 1-2 represent hydraulic fracturing experiments conducted under the same bedding plane dip angle (0°) and viscosity (20 cp) but different injection rates. Under various injection rates, the pressure curves all exhibit distinct stages of fracture initiation, propagation, and extension, as shown in Fig. 12(a). Specimens 4-6 represent experiments conducted with identical injection rates and bedding plane dip angles but different fracturing fluid viscosities, with their pressure curves illustrated in Fig. 12(b).

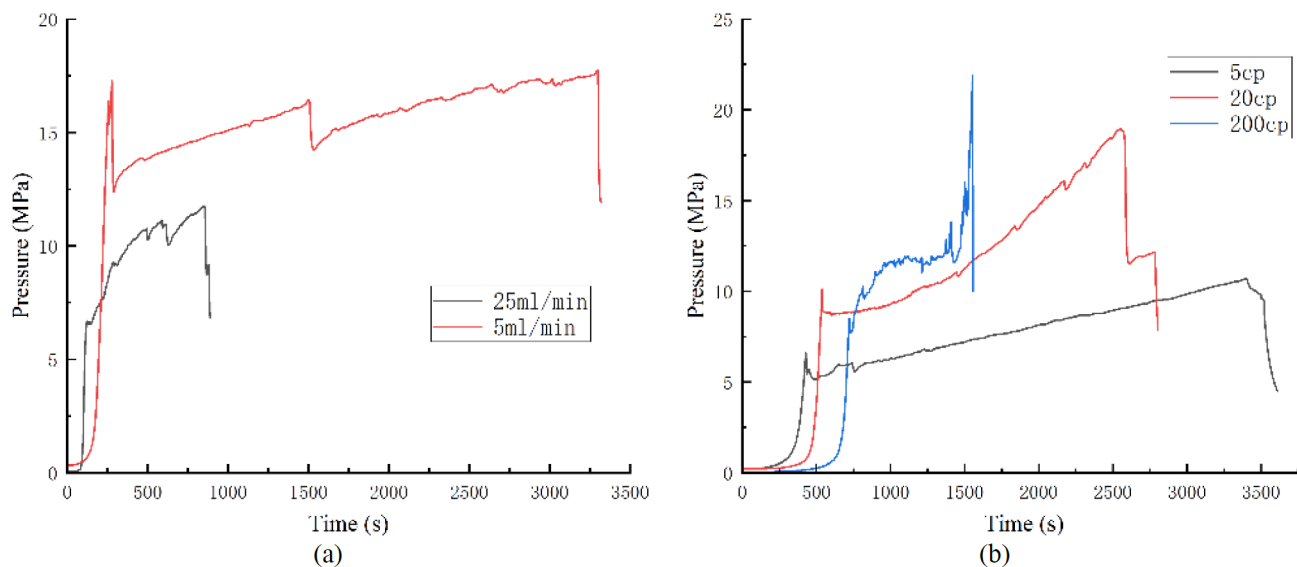


Figure 12—Injection pressure curves under different construction conditions: (a) varying injection rate (b) varying fluid viscosities.

After the experiment, the specimens were sectioned to observe fracture morphologies. A SIMSCAN portable scanner was used to perform 3D scanning of fracture surfaces at different bedding angles. Post-processing of Specimens 1-2 revealed the hydraulic fracture (red) and induced bedding fractures (green) of varying injection rates, as illustrated in Fig. 13. Similarly, post-processing of Specimens 4-6 displayed the hydraulic fracture (red) and induced bedding fractures (green) of varying fluid viscosities, as shown in Fig. 14.

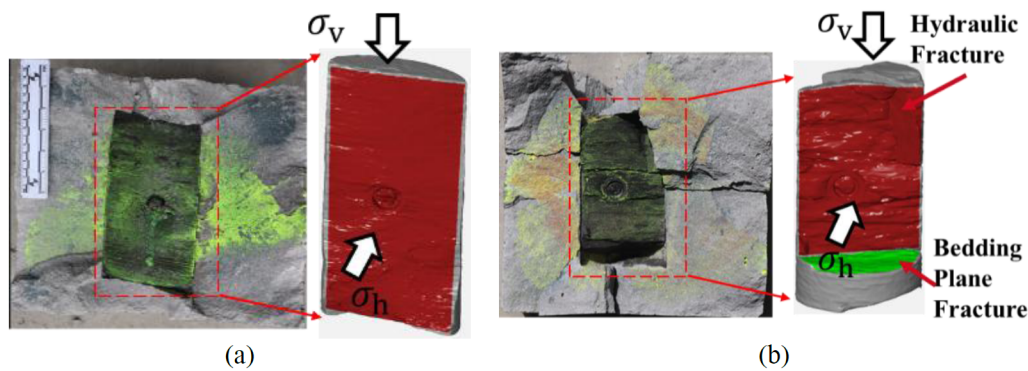


Figure 13—Fracture morphology under different injection rates: (a) 5 ml/min, (b) 25 ml/min

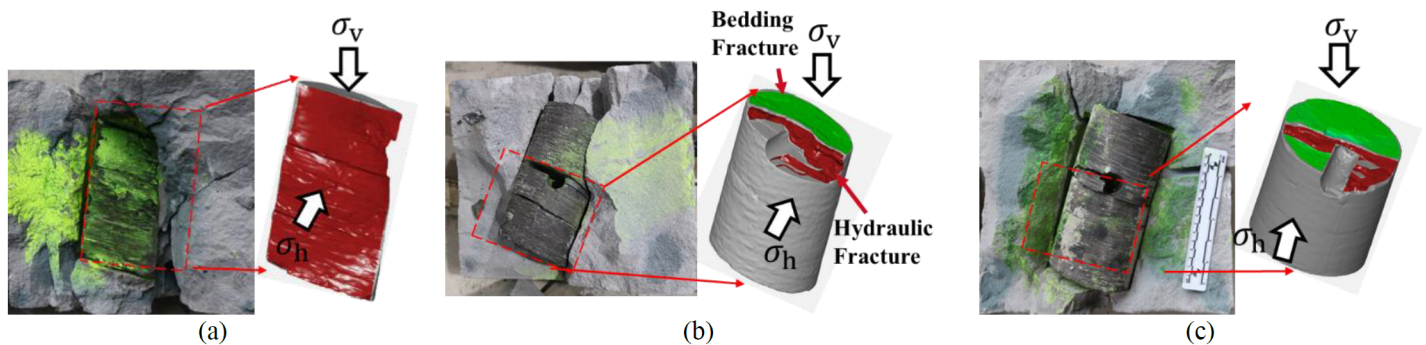


Figure 14—Fracture morphology under different viscosities: (a) 5 cp, (b) 20 cp, (c) 200 cp

As shown in Fig. 8 and 13, natural fractures exhibit a certain influence on hydraulic fractures morphology under different injection rates. At a low injection rate (5 ml/min), the overall fracture surface is relatively smooth, with hydraulic fractures primarily penetrating through layers to form a single straight fracture. At a high injection rate (25 ml/min), the fracture morphology remains similar to that under low injection rates, but surface roughness increases. Hydraulic fractures interact with natural fractures, generating a limited number of bedding fractures and resulting in a more complex fracture surface.

From Fig. 8 and 14, natural fractures have a significant impact on the propagation and morphology of hydraulic fractures under different viscosities. At low viscosity, hydraulic fractures tend to propagate across layers and activate some natural fractures. At medium and high viscosities, the fracture surface morphology is complex and tortuous, with natural fractures opening and hydraulic fractures propagating in a stepped manner along natural fractures.

Based on the fracture surface height under different viscosities and flow rates in Fig. 9, it is found that at the same bedding angle, high viscosity and flow rate cause the failure mechanism of hydraulic fractures to change from tensile failure to tensile-shear failure, but the internal conditions causing local stress changes are different from the bedding angle.

As shown in Fig. 15, when hydraulic fractures intersect with natural fractures, the propagation of hydraulic fractures will be affected by natural fractures. If the fluid pressure at the intersection exceeds the normal stress acting on the natural fractures, the natural fractures will open, resulting in sliding between the fracture surfaces, that is, when the shear stress acting on the fracture surface is greater than the sum of the inherent strength of the fracture surface and the frictional force generated by the effective normal stress acting on the natural fracture surface. When the initiation pressure required by the natural fracture is less than the opening pressure, the hydraulic fracture will propagate forward. Combining with Formula 2 mentioned in Section 3.1, the condition for the sliding of natural fractures under the influence of hydraulic fractures can be obtained as:

$$\tau > (\sigma_n - p)\tan\phi + c \quad (9)$$

Concurrently, the opening of natural fractures requires that:

$$p > \sigma_n \quad (10)$$

Where, p is the net pressure within the hydraulic fracture (MPa); σ_n is the normal stress acting on the natural fracture surface induced by confining pressure (MPa).

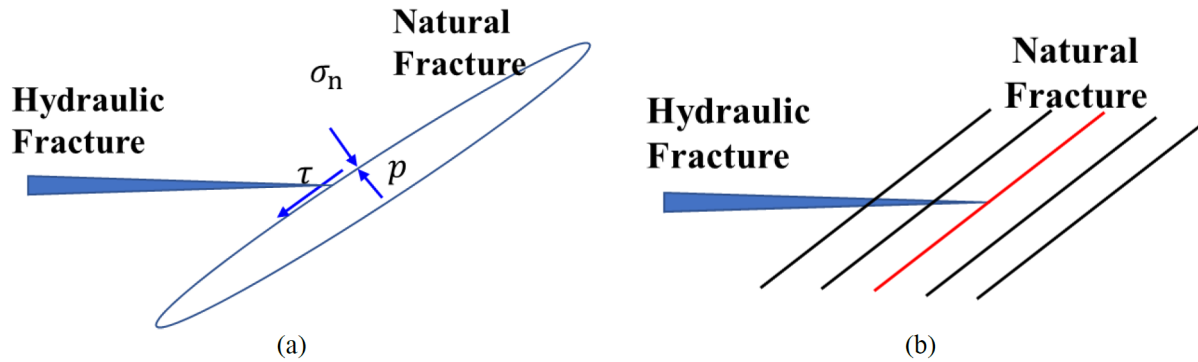


Figure 15—Interaction between hydraulic and natural fractures (a) Stress state at hydraulic fracture tip (b) Activation of natural fractures

In combination with the injection pressure curve in Fig. 12, as viscosity and displacement increase, the net pressure inside the hydraulic fracture increases, the local stress state of the fracture changes, and the shear slip conditions of natural fractures is reduced. When the hydraulic fracture extends to the natural fracture, this weak surface is prone to shear slip and opening, and the hydraulic fracture extends along the natural fracture. However, the in-situ stress remains the dominant factor. The in-situ stress will cause the hydraulic fracture to extend along the direction perpendicular to the minimum principal stress, and the hydraulic fracture will deviate. The failure mechanism of the hydraulic fracture changes from tensile failure to tensile-shear failure until it encounters the next natural fracture surface. The hydraulic fracture extension repeats this process continuously, and thus, under high viscosity and high displacement, the hydraulic fracture extends along the natural fractures in a stepped manner, as shown in Fig. 16.

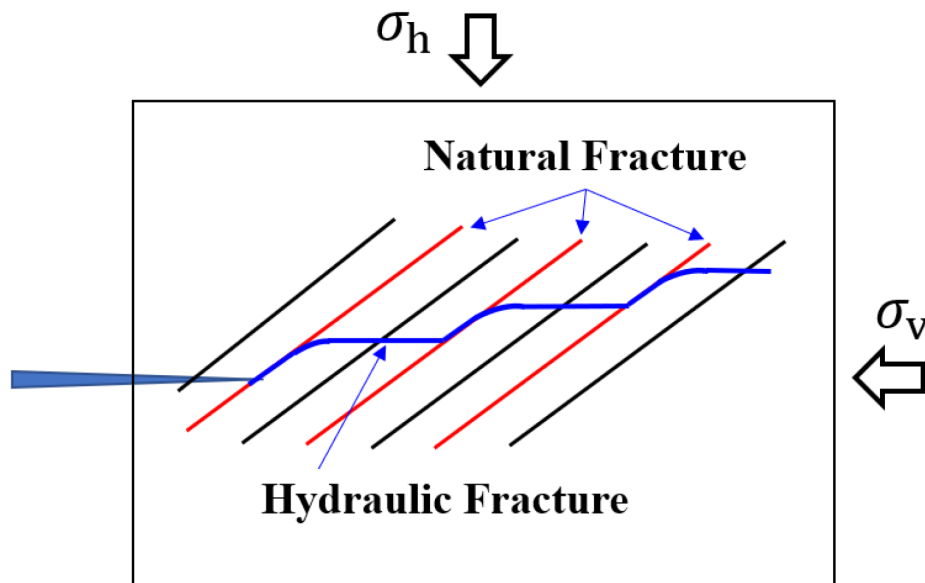


Figure 16—The impact of multiple natural fractures on hydraulic fractures

Conclusions

This study conducted true triaxial hydraulic fracturing experiments on full-diameter shale cores under various bedding plane angles, injection rates, and fracturing fluid viscosities. Three-dimensional scanning techniques were employed to characterize fracture propagation patterns and surface roughness. The main conclusions are as follows:

1. Bedding angles exert a significant influence on the mechanical properties of multi-fractured shale formations. The compressive strength exhibits a tripartite distribution (0-30°, 30-60°, 60-90°), decreasing initially then increasing. At low angles (0-30°), fractures primarily manifest as bedding-parallel + inclined cracks with complex failure surfaces. Medium angles (45-60°) activate bedding plane failures, while high angles (75-90°) result in single shear failure surfaces.
2. Variations in local stress states govern fracture propagation paths. Bedding angles affect shale mechanical properties and stress distribution, while viscosity and injection rate modify net pressure within fractures. These stress variations induce two distinct failure mechanisms: tensile failure and tensile-shear mixed failure.
3. Bedding angles critically control fracture propagation paths and surface characteristics. At 0°, fractures penetrate bedding planes forming straight paths. Increasing angles induce stepwise propagation along bedding interfaces. Fracture roughness and complexity increase with bedding angles, with high-angle surfaces exhibiting 8-fold greater roughness than low-angle counterparts.
4. Fracturing fluid viscosity and injection rate alter fracture morphology and failure mechanisms. The viscosity and flow rate of the high-pressure fracturing fluid increase the net pressure within the fractures, altering the local stress state of the fractures and making natural fractures more likely to be activated. Hydraulic fractures tend to propagate along natural fractures. However, the in-situ stress remains the dominant controlling factor. Hydraulic fractures will expand in a direction perpendicular to the minimum principal stress, and the failure mechanism of hydraulic fractures changes from tensile failure to tensile-shear failure, with the hydraulic fracture expanding in a stepped manner.

Acknowledge

This work was financially supported by the National Science Foundation of China (52434001), the Science Foundation of China University of Petroleum, Beijing (ZX20230085), and the National Science Foundation of China (52274013).

Nomenclature

- c = cohesion of the structural plane (MPa)
- K_I = Mode I fracture toughness values (MPa·m^{1/2})
- K_{II} = Mode II fracture toughness values (MPa·m^{1/2})
- N = the number of analysis points (dimensionless)
- p = net pressure within the hydraulic fracture (MPa)
- r = distance from the crack tip (mm)
- z_{ij} = height of each data analysis point on the fracture surface (mm)
- Z_{rms} = root mean square height of the fracture surface (mm)
- β = the angle between the maximum principal stress and the normal direction of the structural plane (°)
- θ = angle between the crack propagation direction and deflection direction (°)
- σ = principal stresses (MPa)
- σ_1 = first principal stress (MPa)
- σ_3 = third principal stress (MPa)

- σ_n = normal stress acting on the natural fracture surface induced by confining pressure (MPa)
 $\sigma_r, \sigma_\theta, \tau_{r\theta}$ = three principal stresses in cylindrical coordinates (MPa)
 τ = shear stress (MPa)
 φ = internal friction angle ($^\circ$)

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